

# Backend API Specifications for Estimation Workflow System

## API Structure Overview

### Base URL Structure

```
/api/v1/{module}/{resource}/{action}
```

### Authentication & Authorization

All APIs require JWT token except public endpoints

```
javascript
```

```
Headers: {  
  "Authorization": "Bearer <jwt_token>",  
  "Content-Type": "application/json"  
}
```

---

## 1. Authentication & User Management APIs

### AUTH Module

```
POST  /api/v1/auth/login  
POST  /api/v1/auth/logout  
POST  /api/v1/auth/refresh-token  
POST  /api/v1/auth/forgot-password  
POST  /api/v1/auth/reset-password  
GET   /api/v1/auth/me
```

#### Example: Login API

javascript

```
POST /api/v1/auth/login
Body: {
  "email": "john@company.com",
  "password": "password123"
}
Response: {
  "success": true,
  "data": {
    "user": {
      "id": "user_123",
      "name": "John Smith",
      "email": "john@company.com",
      "role": "ESTIMATOR"
    },
    "token": "eyJhbGciOiJIUzI1NiIs... ",
    "refreshToken": "dGhpcyBpcyBhIHJlZnJlc2g..."
  }
}
```

## USER Management APIs

|        |                             |                                    |
|--------|-----------------------------|------------------------------------|
| GET    | /api/v1/users               | // List users (Admin/Manager only) |
| POST   | /api/v1/users               | // Create user (Admin only)        |
| GET    | /api/v1/users/:id           | // Get user details                |
| PUT    | /api/v1/users/:id           | // Update user                     |
| DELETE | /api/v1/users/:id           | // Deactivate user (Admin only)    |
| GET    | /api/v1/users/by-role/:role | // Get users by role               |

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## 2. Client Management APIs

### CLIENT Module

|        |                                    |                                  |
|--------|------------------------------------|----------------------------------|
| GET    | /api/v1/clients                    | // List all clients              |
| POST   | /api/v1/clients                    | // Create new client             |
| GET    | /api/v1/clients/:id                | // Get client details            |
| PUT    | /api/v1/clients/:id                | // Update client                 |
| DELETE | /api/v1/clients/:id                | // Deactivate client             |
| GET    | /api/v1/clients/without-rfq-access | // Clients needing sales support |
| PUT    | /api/v1/clients/:id/rfq-access     | // Toggle RFQ access             |

### Example: Create Client API

javascript

POST /api/v1/clients

```
Body: {
  "name": "John Doe",
  "email": "john@construction.com",
  "phone": "+1-555-0123",
  "company": "ABC Construction",
  "hasRfqAccess": false
}
Response: {
  "success": true,
  "data": {
    "id": "client_456",
    "name": "John Doe",
    "company": "ABC Construction",
    "hasRfqAccess": false,
    "createdAt": "2025-06-19T10:30:00Z"
  }
}
```

---

### 3. RFQ Management APIs

#### RFQ Core APIs

|        |                  |                                 |
|--------|------------------|---------------------------------|
| GET    | /api/v1/rfqs     | // List RFQs (filtered by role) |
| POST   | /api/v1/rfqs     | // Create RFQ                   |
| GET    | /api/v1/rfqs/:id | // Get RFQ details              |
| PUT    | /api/v1/rfqs/:id | // Update RFQ                   |
| DELETE | /api/v1/rfqs/:id | // Delete RFQ (Admin only)      |

#### RFQ Status & Assignment APIs

|      |                                 |                                |
|------|---------------------------------|--------------------------------|
| PUT  | /api/v1/rfqs/:id/status         | // Update RFQ status           |
| POST | /api/v1/rfqs/:id/assign         | // Assign RFQ for estimation   |
| GET  | /api/v1/rfqs/pending-assignment | // RFQs waiting for assignment |
| GET  | /api/v1/rfqs/by-status/:status  | // Filter RFQs by status       |

#### RFQ Document APIs

|        |                                   |                               |
|--------|-----------------------------------|-------------------------------|
| GET    | /api/v1/rfqs/:id/documents        | // List RFQ documents         |
| POST   | /api/v1/rfqs/:id/documents        | // Upload document            |
| GET    | /api/v1/rfqs/:id/documents/:docId | // Download specific document |
| DELETE | /api/v1/rfqs/:id/documents/:docId | // Delete document            |

#### Example: Create RFQ API

```
POST /api/v1/rfqs
Body: {
  "projectName": "Steel Office Building",
  "description": "50-story office complex with parking",
  "clientId": "client_456",
  "salesPersonId": "user_789", // Optional
  "priority": "HIGH",
  "expectedStartDate": "2025-07-01"
}
Response: {
  "success": true,
  "data": {
    "id": "rfq_123",
    "rfqNumber": "RFQ-2025-001",
    "projectName": "Steel Office Building",
    "status": "RECEIVED",
    "client": {
      "name": "John Doe",
      "company": "ABC Construction"
    }
  }
}
```

## 4. Estimation Management APIs

### ESTIMATION Core APIs

|        |                         |                                     |
|--------|-------------------------|-------------------------------------|
| GET    | /api/v1/estimations     | // List estimations (role-filtered) |
| POST   | /api/v1/estimations     | // Create estimation                |
| GET    | /api/v1/estimations/:id | // Get estimation details           |
| PUT    | /api/v1/estimations/:id | // Update estimation                |
| DELETE | /api/v1/estimations/:id | // Delete estimation                |

### ESTIMATION Assignment & Review APIs

|      |                                    |                                            |
|------|------------------------------------|--------------------------------------------|
| POST | /api/v1/estimations/:id/assign     | // Assign to estimator                     |
| PUT  | /api/v1/estimations/:id/review     | // Review estimation (Approve/Reject/Edit) |
| POST | /api/v1/estimations/:id/submit     | // Submit for review                       |
| GET  | /api/v1/estimations/pending-review | // Estimations pending review              |
| GET  | /api/v1/estimations/my-assignments | // User's assigned estimations             |

### ESTIMATION Task APIs

```
GET      /api/v1/estimations/:id/tasks    // List tasks for estimation
POST     /api/v1/estimations/:id/tasks    // Create task (delegate work)
PUT      /api/v1/estimation-tasks/:taskId // Update task
GET      /api/v1/estimation-tasks/:taskId // Get task details
POST     /api/v1/estimation-tasks/:taskId/submit // Submit completed task
```

## ESTIMATION Line Items APIs

```
GET      /api/v1/estimations/:id/line-items    // Get line items
POST     /api/v1/estimations/:id/line-items    // Add line item
PUT      /api/v1/estimation-line-items/:itemId // Update line item
DELETE   /api/v1/estimation-line-items/:itemId // Delete line item
POST     /api/v1/estimations/:id/calculate     // Recalculate totals
```

### Example: Create Estimation API

javascript

```
POST /api/v1/estimations
Body: {
  "rfqId": "rfq_123",
  "fabricatorName": "Steel Works Inc",
  "teklaOrSds": "TEKLA",
  "assignedToId": "user_456", // Estimation Head or Estimator
  "dueDate": "2025-06-30",
  "instructions": "Focus on connection details"
}
Response: {
  "success": true,
  "data": {
    "id": "est_789",
    "estimationNumber": "EST-2025-001",
    "status": "DRAFT",
    "assignedTo": {
      "name": "Senior Estimator",
      "role": "ESTIMATION_HEAD"
    }
  }
}
```

---

## 5. Time Tracking & Progress APIs

### TIME Tracking APIs

```
POST    /api/v1/time-tracking/start      // Start time tracking
POST    /api/v1/time-tracking/stop    // Stop time tracking
POST    /api/v1/time-tracking/break   // Log break time
GET     /api/v1/time-tracking/current  // Get current session
GET     /api/v1/estimation-tasks/:taskId/time-logs // Get time history
```

## Example: Time Tracking API

javascript

```
POST /api/v1/time-tracking/start
Body: {
  "taskId": "task_456",
  "activity": "Line item calculations"
}
Response: {
  "success": true,
  "data": {
    "sessionId": "session_789",
    "startTime": "2025-06-19T09:00:00Z",
    "taskId": "task_456"
  }
}
```

---

## 6. Template & Calculation APIs

### TEMPLATE APIs

```
GET     /api/v1/estimations/:id/template // Get estimation template
PUT     /api/v1/estimations/:id/template // Update template data
POST    /api/v1/templates/generate-email // Generate email from template
```

### CALCULATION APIs

```
POST    /api/v1/calculations/line-item // Calculate line item totals
POST    /api/v1/calculations/estimation // Calculate estimation totals
POST    /api/v1/calculations/formula    // Execute custom formula
GET     /api/v1/calculations/formulas    // Get available formulas
```

## Example: Excel Formula Migration API

javascript

POST /api/v1/calculations/formula

```
Body: {
  "formulaType": "HOURS_PER_TON",
  "inputs": {
    "tonnage": 150,
    "complexity": "MEDIUM",
    "connectionType": "BOLTED"
  }
}
Response: {
  "success": true,
  "data": {
    "result": 8.5,
    "unit": "hours_per_ton",
    "formula": "base_hours * complexity_factor * connection_factor"
  }
}
```

---

## 7. Pricing & Quote APIs

### PRICING APIs

|      |                                 |                            |
|------|---------------------------------|----------------------------|
| GET  | /api/v1/estimations/:id/pricing | // Get pricing details     |
| PUT  | /api/v1/estimations/:id/pricing | // Set pricing (CEO only)  |
| POST | /api/v1/quotes/generate         | // Generate quote document |
| GET  | /api/v1/quotes/:id              | // Get quote details       |
| POST | /api/v1/quotes/:id/send         | // Send quote to client    |

### Example: Set Pricing API

javascript

PUT /api/v1/estimations/est\_789/pricing

```
Body: {
  "hourlyRate": 125,
  "detailingPrice": 42500,
  "mainSteelConnectionPrice": 15000,
  "miscSteelDesignPrice": 8500,
  "profitMargin": 22,
  "validityPeriod": 30,
  "paymentTerms": "Net 30"
}
Response: {
  "success": true,
  "data": {
    "finalQuoteAmount": 66000,
    "quoteId": "quote_123",
    "status": "READY_TO_SEND"
  }
}
```

---

## 8. Project Management APIs (For Awarded Projects)

### PROJECT APIs

|      |                             |                                      |
|------|-----------------------------|--------------------------------------|
| GET  | /api/v1/projects            | // List active projects              |
| POST | /api/v1/projects            | // Create project (from awarded RFQ) |
| GET  | /api/v1/projects/:id        | // Get project details               |
| PUT  | /api/v1/projects/:id        | // Update project                    |
| PUT  | /api/v1/projects/:id/status | // Update project status             |

### PROJECT Document APIs

|        |                                       |                            |
|--------|---------------------------------------|----------------------------|
| GET    | /api/v1/projects/:id/documents        | // List project documents  |
| POST   | /api/v1/projects/:id/documents        | // Upload project document |
| GET    | /api/v1/projects/:id/documents/:docId | // Download document       |
| DELETE | /api/v1/projects/:id/documents/:docId | // Delete document         |

---

## 9. Analytics & Reporting APIs

### DASHBOARD APIs

|     |                                   |                               |
|-----|-----------------------------------|-------------------------------|
| GET | /api/v1/dashboard/overview        | // Role-based dashboard data  |
| GET | /api/v1/dashboard/metrics         | // Key performance indicators |
| GET | /api/v1/dashboard/recent-activity | // Recent activities          |

## ANALYTICS APIs

|     |                                        |                          |
|-----|----------------------------------------|--------------------------|
| GET | /api/v1/analytics/estimator-efficiency | // Individual efficiency |
| GET | /api/v1/analytics/project-success-rate | // Win/loss rates        |
| GET | /api/v1/analytics/estimation-accuracy  | // Estimated vs actual   |
| GET | /api/v1/analytics/pipeline-value       | // Revenue pipeline      |
| GET | /api/v1/analytics/team-performance     | // Team metrics          |

## REPORTS APIs

|      |                                  |                                |
|------|----------------------------------|--------------------------------|
| GET  | /api/v1/reports/monthly-summary  | // Monthly business summary    |
| GET  | /api/v1/reports/estimator-report | // Individual estimator report |
| GET  | /api/v1/reports/project-timeline | // Project timeline analysis   |
| POST | /api/v1/reports/custom           | // Custom report generation    |
| GET  | /api/v1/reports/export/:reportId | // Export report (PDF/Excel)   |

## Example: Dashboard Overview API

javascript

GET /api/v1/dashboard/overview

```
Response: {
  "success": true,
  "data": {
    "userRole": "DEPT_MANAGER",
    "pendingAssignment": 5,
    "activeEstimations": 12,
    "pendingReview": 3,
    "teamWorkload": [
      {
        "estimatorName": "John Smith",
        "activeTasks": 3,
        "efficiency": "92%"
      }
    ],
    "recentActivity": [
      {
        "type": "ESTIMATION_COMPLETED",
        "project": "Office Complex",
        "estimator": "Jane Doe",
        "timestamp": "2025-06-19T08:30:00Z"
      }
    ]
  }
}
```

## NOTIFICATION APIs

|        |                                |                               |
|--------|--------------------------------|-------------------------------|
| GET    | /api/v1/notifications          | // Get user notifications     |
| PUT    | /api/v1/notifications/:id/read | // Mark notification as read  |
| POST   | /api/v1/notifications/send     | // Send notification (system) |
| DELETE | /api/v1/notifications/:id      | // Delete notification        |

## COMMUNICATION APIs

|      |                                  |                              |
|------|----------------------------------|------------------------------|
| POST | /api/v1/communications/email     | // Send email                |
| GET  | /api/v1/communications/templates | // Email templates           |
| POST | /api/v1/communications/comment   | // Add comment to estimation |
| GET  | /api/v1/estimations/:id/comments | // Get estimation comments   |

---

## 11. File Management APIs

### FILE Upload/Download APIs

|        |                            |                         |
|--------|----------------------------|-------------------------|
| POST   | /api/v1/files/upload       | // Generic file upload  |
| GET    | /api/v1/files/:fileId      | // Download file        |
| DELETE | /api/v1/files/:fileId      | // Delete file          |
| GET    | /api/v1/files/:fileId/info | // Get file metadata    |
| POST   | /api/v1/files/batch-upload | // Multiple file upload |

### DOCUMENT Processing APIs

|      |                                |                                |
|------|--------------------------------|--------------------------------|
| POST | /api/v1/documents/extract-text | // Extract text from PDF/image |
| POST | /api/v1/documents/convert      | // Convert document format     |
| GET  | /api/v1/documents/:id/preview  | // Generate document preview   |

---

## 12. History & Audit APIs

### HISTORY Tracking APIs

|     |                                 |                              |
|-----|---------------------------------|------------------------------|
| GET | /api/v1/estimations/:id/history | // Estimation change history |
| GET | /api/v1/rfqs/:id/history        | // RFQ status history        |
| GET | /api/v1/users/:id/activity      | // User activity log         |
| GET | /api/v1/audit/changes           | // System-wide audit log     |

---

## API Response Standards

### Success Response Format

javascript

```
{
  "success": true,
  "data": { /* response data */ },
  "message": "Optional success message",
  "pagination": { // For list endpoints
    "page": 1,
    "limit": 20,
    "total": 150,
    "pages": 8
  }
}
```

## Error Response Format

javascript

```
{
  "success": false,
  "error": {
    "code": "VALIDATION_ERROR",
    "message": "Invalid input data",
    "details": [
      {
        "field": "email",
        "message": "Email is required"
      }
    ]
  }
}
```

## HTTP Status Codes

- **200:** Success
- **201:** Created
- **400:** Bad Request (validation errors)
- **401:** Unauthorized
- **403:** Forbidden (role access)
- **404:** Not Found
- **409:** Conflict (duplicate data)
- **500:** Internal Server Error

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## Security & Middleware

## Required Middleware for All APIs

1. **Authentication Middleware:** Verify JWT token
2. **Authorization Middleware:** Check role permissions
3. **Rate Limiting:** Prevent API abuse
4. **Request Validation:** Validate input data
5. **Audit Logging:** Log all API calls
6. **CORS:** Configure cross-origin requests

## Role-Based API Access

javascript

```
// Example middleware setup
const rolePermissions = {
  'ADMIN': ['*'], // All APIs
  'DEPT_MANAGER': ['rfqs.*', 'estimations.*', 'users.read', 'analytics.*'],
  'ESTIMATION_HEAD': ['estimations.*', 'estimation-tasks.*'],
  'ESTIMATOR': ['estimation-tasks.assigned', 'time-tracking.*'],
  'SALES_PERSON': ['rfqs.create', 'clients.*'],
  'CLIENT': ['rfqs.own', 'quotes.own']
};
```

This comprehensive API specification covers all the functionality needed for your estimation workflow system while maintaining security, scalability, and clear separation of concerns.